


CLASSROOM CONTACT PROGRAMME

(Academic Session : 2023 - 2024)

Test Pattern

JEE(Advanced)
FULL SYLLABUS
 24-04-2024

JEE(Main + Advanced) : ENTHUSIAST COURSE ALL STAR BATCH (SCORE ADVANCED)

Time : 3 Hours

PAPER-2 (OPTIONAL)

Maximum Marks : 186

READ THE INSTRUCTIONS CAREFULLY
GENERAL :

1. This sealed booklet is your Question Paper. Do not break the seal till you are told to do so.
2. Use the Optical Response sheet (ORS) provided separately for answering the questions.
3. Blank spaces are provided within this booklet for rough work.
4. Write your name, form number and sign in the space provided on the back cover of this booklet.
5. After breaking the seal of the booklet, verify that the booklet contains **24** pages and that all the **18** questions in each subject and along with the options are legible. If not, contact the invigilator for replacement of the booklet.
6. You are allowed to take away the Question Paper at the end of the examination.

OPTICAL RESPONSE SHEET :

7. The ORS will be collected by the invigilator at the end of the examination.
8. Do not tamper with or mutilate the ORS. **Do not use the ORS for rough work.**
9. Write your name, form number and sign with pen in the space provided for this purpose on the ORS. **Do not write any of these details anywhere else on the ORS.** Darken the appropriate bubble under each digit of your form number.

DARKENING THE BUBBLES ON THE ORS :

10. Use a **BLACK BALL POINT PEN** to darken the bubbles on the ORS.
11. Darken the bubble  **COMPLETELY.**
12. The correct way of darkening a bubble is as : 
13. The ORS is machine-gradable. Ensure that the bubbles are darkened in the correct way.
14. Darken the bubbles **ONLY IF** you are sure of the answer. There is **NO WAY** to erase or "un-darken" a darkened bubble.
15. Take **$g = 10 \text{ m/s}^2$** unless otherwise stated.

QUESTION PAPER FORMAT :

16. The question paper has three parts : Physics, Chemistry and Mathematics.

Please see the last page of this booklet for rest of the instructions

DO NOT BREAK THE SEALS WITHOUT BEING INSTRUCTED TO DO SO BY THE INVIGILATOR

SOME USEFUL CONSTANTS

Atomic No. : H = 1, B = 5, C = 6, N = 7, O = 8, F = 9, Al = 13, P = 15, S = 16, Cl = 17, Br = 35, Xe = 54, Ce = 58

Atomic masses : H = 1, Li = 7, B = 11, C = 12, N = 14, O = 16, F = 19, Na = 23, Mg = 24, Al = 27, P = 31, S = 32, Cl = 35.5, Ca = 40, Fe = 56, Br = 80, I = 127, Xe = 131, Ba = 137, Ce = 140

- | | |
|------------------------------------|--|
| • Boltzmann constant | $k = 1.38 \times 10^{-23} \text{ JK}^{-1}$ |
| • Coulomb's law constant | $\frac{1}{4\pi\epsilon_0} = 9 \times 10^9$ |
| • Universal gravitational constant | $G = 6.67259 \times 10^{-11} \text{ N-m}^2 \text{ kg}^{-2}$ |
| • Speed of light in vacuum | $c = 3 \times 10^8 \text{ ms}^{-1}$ |
| • Stefan-Boltzmann constant | $\sigma = 5.67 \times 10^{-8} \text{ Wm}^{-2}\text{-K}^{-4}$ |
| • Wien's displacement law constant | $b = 2.89 \times 10^{-3} \text{ m-K}$ |
| • Permeability of vacuum | $\mu_0 = 4\pi \times 10^{-7} \text{ NA}^{-2}$ |
| • Permittivity of vacuum | $\epsilon_0 = \frac{1}{\mu_0 c^2}$ |
| • Planck constant | $h = 6.63 \times 10^{-34} \text{ J-s}$ |

Space for Rough Work

PART-1 : PHYSICS

SECTION-I (i) : (Maximum Marks: 32)

- This section contains **EIGHT (08)** questions.
- Each question has **FOUR** options. **ONE OR MORE THAN ONE** of these four option(s) is (are) correct answer(s).
- For each question, choose the option(s) corresponding to (all) the correct answer(s)
- Answer to each question will be evaluated according to the following marking scheme:

Full Marks : +4 If only (all) the correct option(s) is (are) chosen.

Partial Marks : +3 If all the four options are correct but **ONLY** three options are chosen.

Partial Marks : +2 If three or more options are correct but **ONLY** two options are chosen and both of which are correct.

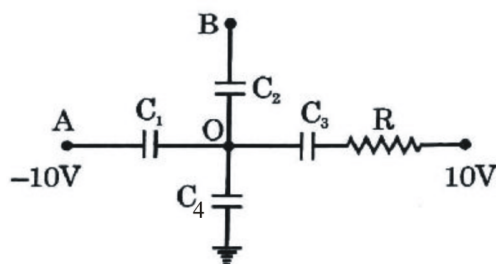
Partial Marks : +1 If two or more options are correct but **ONLY** one option is chosen and it is a correct option.

Zero Marks : 0 If none of the options is chosen (i.e. the question is unanswered).

Negative Marks : -1 In all other cases.

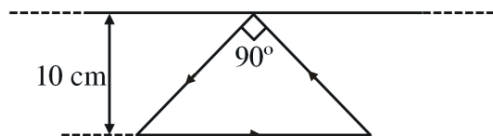
- **For Example** : If first, third and fourth are the **ONLY** three correct options for a question with second option being an incorrect option; selecting only all the three correct options will result in +4 marks. Selecting only two of the three correct options (e.g. the first and fourth options), without selecting any incorrect option (second option in this case), will result in +2 marks. Selecting only one of the three correct options (either first or third or fourth option), without selecting any incorrect option (second option in this case), will result in +1 marks. Selecting any incorrect option(s) (second option in this case), with or without selection of any correct option(s) will result in -1 marks.

1. Figure shows small part of circuit. [Given $C_1 = C_2 = C_3 = C_4 = 5 \mu\text{F}$] which of the following statement is/are TRUE? (Assume steady state)

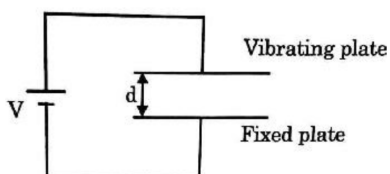


- (A) Potential of point O can be determined but that of B can't be determined
- (B) Potential of both point O & B can't be determined
- (C) If charge on capacitor C_2 were also specified, potential of point O can be determined but that of B can't be determined
- (D) If charge on capacitor C_2 were also specified, potential of both the point O & B can be determined

2. A conducting loop in the shape of a right angled isosceles triangle of height 10 cm is kept such that the 90° vertex is very close to an infinitely long conducting wire (see the figure). The wire is electrically insulated from the loop. The hypotenuse of the triangle is parallel to the wire. The current in the triangular loop is in counterclockwise direction and increased at a constant rate of 10 As^{-1} . Which of the following statement(s) is (are) true?

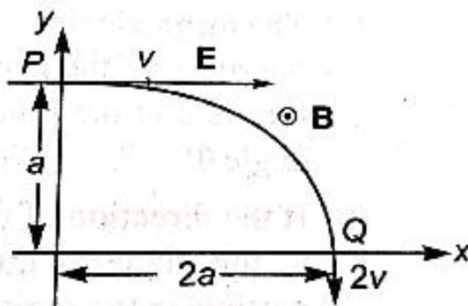


- (A) There is a repulsive force between the wire and the loop
- (B) If the loop is rotated at a constant angular speed about the wire, an additional emf of $\left(\frac{\mu_0}{\pi}\right)$ volt is induced in the wire.
- (C) The magnitude of induced emf in the wire is $\left(\frac{\mu_0}{\pi}\right)$ volts.
- (D) The induced current in the wire is in opposite direction to the current along the hypotenuse.
3. Assume that the earth changes its shape and turns into an infinite cylinder whose radius and distance of moon from the axis of cylindrical earth remains unchanged and also moon remains spherical. Which of the following statement(s) is/are CORRECT?
- (A) Gravitational field at the moon due to earth increases
- (B) Gravitational field at the moon due to earth decreases
- (C) Speed of the moon in its orbit around the cylindrical earth increases
- (D) Speed of the moon in its orbit around the cylindrical earth decreases
4. An air-filled parallel plate capacitor with the plate area A is connected to a battery with an emf V volt and negligible internal resistance. One of the plates vibrates so that the distance between plates varies as $d = d_0 + \alpha \cos \omega t$ (where $\alpha \ll d_0$). The capacitor breaks down when the instantaneous current reaches to the value ' I '. Choose the CORRECT option(s)

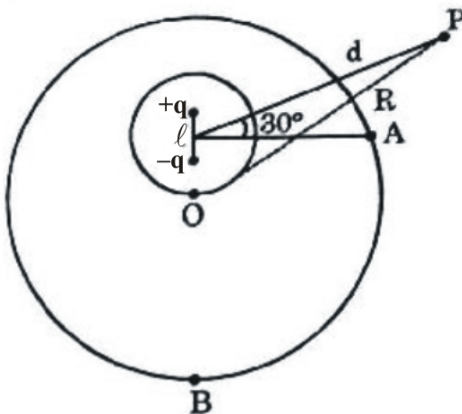


- (A) The maximum possible amplitude vibrations ' a ' is $\frac{Id_0^2}{V A \omega \epsilon_0}$
- (B) The maximum possible amplitude vibrations ' a ' is $\frac{2Id_0^2}{V A \omega \epsilon_0}$
- (C) The instantaneous current in circuit is $\frac{\epsilon_0 V A \omega \alpha}{(d_0 + \alpha \cos \omega t)^2} \sin \omega t$
- (D) The instantaneous current in circuit is $\frac{\epsilon_0 V A \omega \alpha}{2(d_0 + \alpha \cos \omega t)^2} \sin \omega t$

5. A particle of charge $+q$ and mass ' m ' moving under the influence of a uniform electric field $E\hat{i}$ and uniform magnetic field $B\hat{k}$ follows a trajectory from P to Q as shown in figure. The velocities at P and Q are $v\hat{i}$ and $-2v\hat{j}$. Which of the following statement(s) is/are correct?

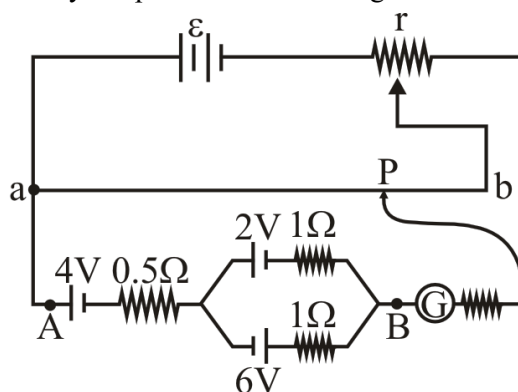


- (A) $E = \frac{3}{4} \left[\frac{mv^2}{qa} \right]$
- (B) Rate of work done by the electric field at P is $\frac{3}{4} \left[\frac{mv^3}{a} \right]$
- (C) Rate of work done by the electric field at P is zero
- (D) Rate of work done by both the fields at Q is zero
6. A cavity is taken out from a uniform conducting sphere. Inside the cavity a dipole is placed as shown in the figure. Find which of the following is/are true
(Given: $q = 10^{-1} \text{C}$, $\ell = 0.1 \text{ mm}$, $\theta = 30^\circ$, $d = 10 \text{ cm}$, $R = 12 \text{ cm}$)



- (A) Potential at points A & B are same
- (B) Potential at points A & B are different
- (C) Potential at point P is 0 volt
- (D) Potential at point P is 2 volt

7. Water jet coming out of a stationary horizontal tube at speed $v = 2\text{ m/s}$ strikes horizontally a massive wall moving in opposite direction with same speed. Water comes to rest relative to wall after striking. Cross-section of jet is 3 cm^2 and density of water is 1 gm/cc . Choose the CORRECT statement(s)
- (A) Force exerted on the wall is 1.2 N
- (B) Force exerted on the wall is 2.4 N
- (C) Rate of change of kinetic energy of water jet striking the wall is 19.2 J/s
- (D) Rate of change of kinetic energy of water jet striking the wall is zero
8. A potentiometer as shown in figure in which three batteries between A & B are connected when the point P is the null point, measured by the potentiometer arrangement. Then choose correct statement(s)



- (A) Current passing through 6 V battery is 4 A
- (B) Current passing through 2 V battery is 4 A
- (C) $V_A - V_B = 2\text{ V}$
- (D) $V_A - V_B = 6\text{ V}$

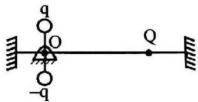
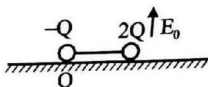
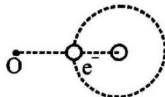
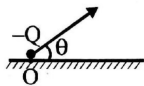
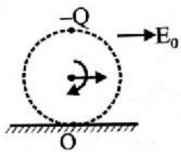
SECTION-I (ii) : (Maximum Marks: 12)

- This section contains **TWO (02)** List-Match sets.
- Each List-Match set has **TWO** Multiple Choice Questions.
- Each List-Match set has two lists : **List-I** and **List-II**
- **List-I** has Four entries (I), (II), (III) and (IV) and **List-II** has **Six** entries (P), (Q), (R), (S), (T) and (U)
- **Four** options are given in each Multiple Choice Question based on List-I and List-II and **ONLY ONE** of these four options satisfies the condition asked in the Multiple Choice Question.
- Answer to each question will be evaluated according to the following marking scheme :

Full Marks : +3 If **ONLY** the option corresponding to the correct combination is chosen.
Zero Marks : 0 If none of the options is chosen (i.e., the question is unanswered)
Negative Marks : -1 In all other cases

Paragraph for Questions 9 and 10

Answer Q.9 and Q.10 by appropriately matching the lists based on the information given in the paragraph.

	List-I		List-II
(I)	Magnitude of angular momentum of system about O increases after the instant shown	(P)	A dipole is hinged on a long smooth fixed horizontal wire. A charge Q (A) is threaded on wire and is free to move on the wire. The system comprises of the dipole and point charge. Gravity is absent 
(II)	The net torque on body A is zero about O at the instant shown	(Q)	A system of two charges $-Q$ and $2Q$ (A) of mass m each are connected by a light rod and kept on smooth level ground. An external uniform electric field $E_0 = mg/Q$ is applied as shown. Gravity is present 
(III)	The net force on negative charge increases in magnitude after the instant shown	(R)	An electron (A) is revolving about a fixed proton in anticlockwise direction. The system comprises of the electron and proton. Gravity is absent 
(IV)	Mechanical energy of the system increases after the instant shown (ignore potential energy of external electrical field)	(S)	System consists of a particle (A) of mass m and charge $-Q$. It is projected from level ground in presence of constant electric field $E = mg/Q$ in vertically upward direction. Gravity is present 
		(T)	System consists of a uniform ring and particle (A) of mass m and charge $-Q$ on its circumference as shown. The ring is in pure rolling on a sufficiently rough horizontal surface. Gravity is present and E_0 is external electric field 
		(U)	None of these

9. Choose correct alternative for the above list

(A) I-P,Q,S ; II-P,R,S

(B) I-P,Q,R ; II-P,R

(C) I-P,Q ; II-P,R,S

(D) I-P,Q,R ; II-P,R,S

10. Choose correct alternative for the above list

(A) III-P, Q,T ; IV-P,Q

(B) III-P,Q ; IV-Q

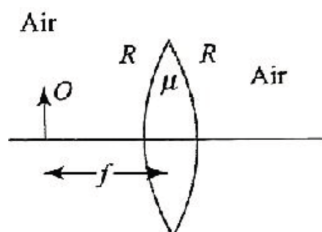
(C) III-P,Q,R ; IV-P

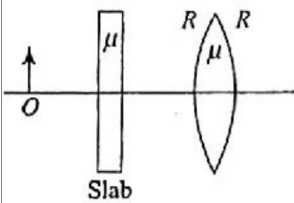
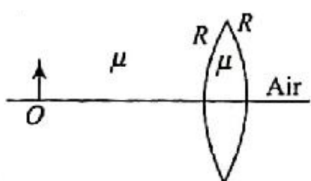
(D) III-P, Q ; IV-P

Paragraph for Questions 11 and 12

Answer Q.11 and Q.12 by appropriately matching the lists based on the information given in the paragraph.

An object O (real) is placed at focus of an equi-biconvex lens as shown in figure. The refractive index of the lens is $\mu = 1.5$ and the radius of curvature of either surface of lens is R . The lens is surrounded by air. In each statement of List-I, some changes are made to situation given above and information regarding final image formed as a result is given in List-II. The distance between lens and object is unchanged in all statements of List-I. Match the statements in List-I with resulting image in List-II.



	List-I		List-II
(I)	If the refractive index of the lens is doubled (that is, made 2μ), then	(P)	Final image is real
(II)	If the radius of curvature is doubled (that is, made $2R$), then	(Q)	Final image is virtual
(III)	If a glass slab of refractive index $\mu = 1.5$ is introduced between the object and lens as shown, then 	(R)	Final image is always enlarged.
(IV)	If the left side of lens is filled with a medium of refractive index $\mu = 1.5$ as shown, then 	(S)	Final image is of same size as the object
		(T)	Final image becomes smaller in size in comparison to size of image before the change was made
		(U)	Image formation is not possible.

11. Choose correct options for conditions in I and II

(A) I-P; II-P,Q,R

(B) I-P,T; II-P,Q,T

(C) I-P,T; II- Q,T

(D) I-P; II-P,Q,S

12. Choose correct options for conditions in III and IV

(A) III-Q,T; IV-Q,T

(B) III-P,T; IV-Q,T

(C) III-P; IV- P,Q,R

(D) III-Q,T; IV-P,Q,T

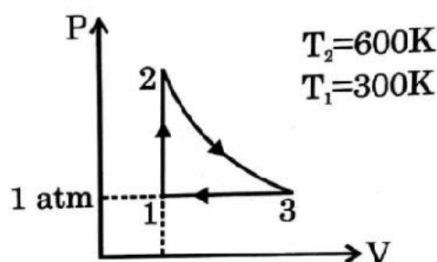
SECTION-II : (Maximum Marks: 18)

- This section contains **SIX (06)** questions. The answer to each question is a **NUMERICAL VALUE**.
- For each question, enter the correct numerical value of the answer in the place designated to enter the answer. If the numerical value has more than two decimal places, **truncate/round-off** the value to **Two** decimal places; e.g. 6.25, 7.00, -0.33, -0.30, 30.27, -127.30, if answer is 11.36777..... then both 11.36 and 11.37 will be correct)
- Answer to each question will be evaluated according to the following marking scheme:

Full Marks : +3 If ONLY the correct numerical value is entered.

Zero Marks : 0 In all other cases.

1. A source of sound with natural frequency f_0 moves uniformly along a straight line. A stationary sound detector is located at a distance of 300 m from the straight line. When the source is closest to the detector, it records a frequency of $3 f_0$ as received from the source. Find the distance (in 'm') of the source from the detector at the instant when it record natural frequency f_0 of the source. If your answer is $300\sqrt{\frac{x}{y}}$ where x & y are reduces to nearest integers from 1 to 9 then fill value (x) + (y).
2. The drag force on a spherical CO_2 bubble in water is given by $F_0 = 6\pi\eta rv$, where η is the viscosity of the surrounding fluid and r is the radius of the sphere (Their relation is called stoke's law). Bubble can be consider to be almost a mass less sphere of radius r moving up in a liquid with terminal speed. Ignore any expansion of bubble during rise. Calculate time taken by sphere to travel distance 20 cm in seconds. (Given; Radius $r = 3 \text{ mm}$, density of water $\rho = 10^3 \text{ kg/m}^3$, Density of $\text{CO}_2 = 1 \text{ kg/m}^3$, $g = 10 \text{ m/s}^2$, $\eta = 0.5 \text{ SI unit}$)
3. We would like to increase the length of a 15 cm long copper rod of cross-section 4 mm^2 by 1 mm. The energy absorbed by the rod if it is heated is E_1 . The energy stored by the rod if it is stretched slowly is E_2 . Then find E_1 / E_2 . If your answer is x fill value of x/100.
[Various parameters of Copper are : Density = $9 \times 10^3 \text{ kg/m}^3$, Thermal coefficient of linear expansion = $8 \times 10^{-5} \text{ K}^{-1}$, Young's modulus = $135 \times 10^9 \text{ Pa}$, Specific heat = 400 J / kg - K]
4. A heat engine takes 0.4 moles of an ideal gas around the cycle as shown in the figure. Process $1 \rightarrow 2$ is isochoric, process $2 \rightarrow 3$ is adiabatic and process $3 \rightarrow 1$ is isobaric. The value of γ for the gas is $5/3$. Find the net heat flow during the cycle in joule. If your answer is N fill value of N/50 ?
[Take: Universal gas constant (R) = $50 / 6 \text{ J mol}^{-1} \text{ K}^{-1}$: $1 \text{ atm} = 10^6 \text{ N/m}^2$ and $2^{3/5} = 1.5$]



5. A certain planet of radius R is composed of a uniform material which generates net power P through radioactive decay. The heat conducted per second $\frac{\Delta Q}{\Delta t} = P \frac{r^3}{R^3}$, where r is radial distance from centre. This results in a temperature differential between the inside and outside of the planet as heat is transferred from the interior to the surface. The rate of heat transfer inside is governed by conduction. It is found that thermal conductivity k is constant for the planet. Assume that the planet is in a steady state i.e., temperature depends on position, but does not depend on time. (Assuming black body radiation and emissivity is 1). If T_C and T_S are temperatures at center and surface respectively. Then temperature difference $T_C - T_S = \frac{\sigma R^m T_S^n}{\ell k}$ fill value of $(m + n + \ell)$
6. When an α -particle collides with a stationary lithium nucleus, the following nuclear reaction may Take place ${}^4\text{He} + {}^7\text{Li} \rightarrow {}^{10}\text{B} + n$ if Q -value of this reaction is $E = -2.8$ MeV, find the minimum Value of kinetic energy (in MeV) of incoming α -particles required for the reaction.

PART-2 : CHEMISTRY
SECTION-I (i) : (Maximum Marks: 32)

- This section contains **EIGHT (08)** questions.
- Each question has **FOUR** options. **ONE OR MORE THAN ONE** of these four option(s) is (are) correct answer(s).
- For each question, choose the option(s) corresponding to (all) the correct answer(s)
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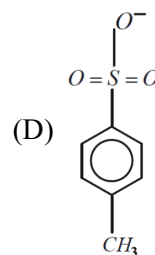
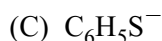
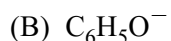
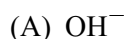
Partial Marks : +1 If two or more options are correct but **ONLY** one option is chosen and it is a correct option.

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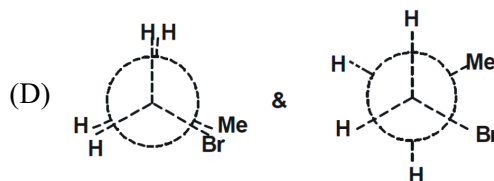
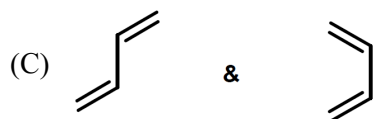
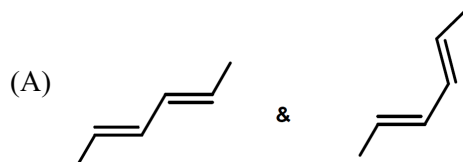
Negative Marks : -1 In all other cases.

- **For Example :** If first, third and fourth are the **ONLY** three correct options for a question with second option being an incorrect option; selecting only all the three correct options will result in +4 marks. Selecting only two of the three correct options (e.g. the first and fourth options), without selecting any incorrect option (second option in this case), will result in +2 marks. Selecting only one of the three correct options (either first or third or fourth option), without selecting any incorrect option (second option in this case), will result in +1 marks. Selecting any incorrect option(s) (second option in this case), with or without selection of any correct option(s) will result in -1 marks.

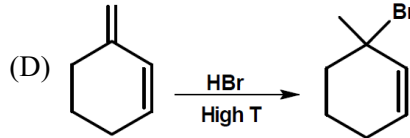
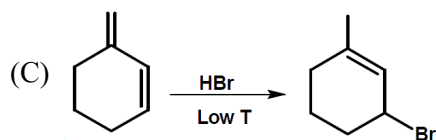
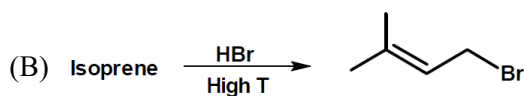
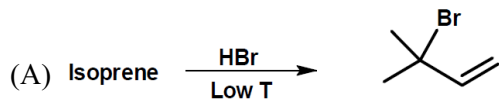
1. In polar protic solvent, among the following which is/are strongly nucleophilic than acetate ion



2. Which of the following is not a correct conformer pair?



3. In parallel path reactions, product from stable intermediate is called K.C.P and it is major product at low temperature. The stable product is called T.C.P and it is major product at high temperature. In which of the following reactions products mentioned are correct ?



4. With which of the following reagents, ferrous ions give red colouration in the presence of ferric ions
- (A) KSCN (B) $\text{NH}_3 + \text{dmg}$
 (C) O-Phenanthroline (D) 2, 2'-Bipyridyl
5. Which of the following molecular orbitals is/are gerade
- (A) σ_{p-p} (B) σ_{p-p}^* (C) δ_{d-d} (D) π_{p-p}^*
6. Which of the following are soluble in CS_2 ?
- (A) White phosphorus (B) Red phosphorus
 (C) Orthorhombic sulphur (D) Monoclinic sulphur
7. During the extraction of copper from copper pyrites, in the Bessemer convertor which of the following reactions take place
- (A) $\text{FeO} + \text{SiO}_2 \rightarrow \text{FeSiO}_3$ (B) $\text{Cu}_2\text{S} + 2\text{Cu}_2\text{O} \rightarrow \text{Cu} + \text{SO}_2$
 (C) $\text{Cu}_2\text{S} + \text{O}_2 \rightarrow \text{Cu}_2\text{O} + \text{SO}_2$ (D) $\text{C} + \text{O}_2 \rightarrow \text{CO}_2$
8. 10 ml mixture containing 0.1 M NaHCO_3 and 0.1M Na_2CO_3 is titrated against an acid with an acid–base indicator. Among the following which observations is/are correct
- (A) When phenolphthalene indicator used, end point observed when 10ml of 1M HCl is added
 (B) After getting end point with phenolphthalein if we added methyl orange indicator, to get next end point 20 ml of 0.1M HCl is to be added furtherly
 (C) When methyl orange indicator used from the beginning, end point observed when 6ml of 0.5M HCl added
 (D) When phenolphthalene indicator used, gram equivalent weight of Na_2CO_3 is 106 g but when methyl orange indicator used gram equivalent weight of Na_2CO_3 is 53g

SECTION-I (ii) : (Maximum Marks: 12)

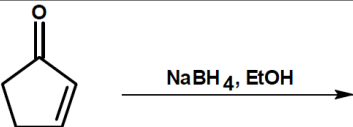
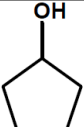
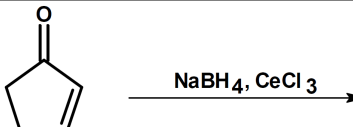
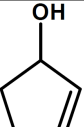
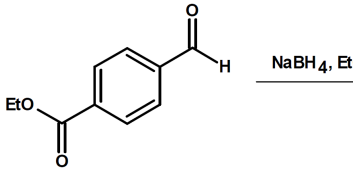
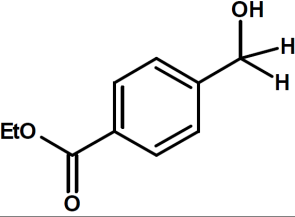
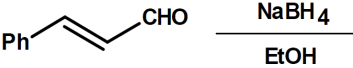
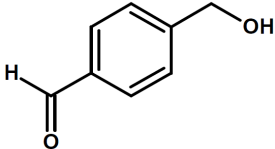
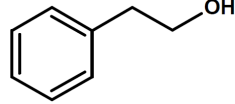
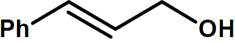
- This section contains **TWO (02)** List-Match sets.
- Each List-Match set has **TWO** Multiple Choice Questions.
- Each List-Match set has two lists : **List-I** and **List-II**
- **List-I** has Four entries (I), (II), (III) and (IV) and **List-II** has **Six** entries (P), (Q), (R), (S), (T) and (U)
- **Four** options are given in each Multiple Choice Question based on List-I and List-II and **ONLY ONE** of these four options satisfies the condition asked in the Multiple Choice Question.
- Answer to each question will be evaluated according to the following marking scheme :

Full Marks : +3 If ONLY the option corresponding to the correct combination is chosen.
 Zero Marks : 0 If none of the options is chosen (i.e., the question is unanswered)
 Negative Marks : -1 In all other cases

Paragraph for Questions 9 and 10

Answer Q.9 and Q.10 by appropriately matching the lists based on the information given in the paragraph.

List-I includes starting materials and reagents. List-II gives the structures of product compounds from the reactions of List-I.

	List-I		List-II
(I)		(P)	
(II)		(Q)	
(III)		(R)	
(IV)		(S)	
		(T)	
		(U)	

9. Which of the following has correct combination considering List – I and List – II ?

- (A) I-P (B) I-Q (C) II-P (D) III-S

10. Which of the following has correct combination considering List – I and List – II ?

- (A) III – R (B) III – S (C) IV – T (D) II – P

Paragraph for Questions 11 and 12

Answer Q.11 and Q.12 by appropriately matching the lists based on the information given in the paragraph.

List-I includes various thermodynamic processes involved for n moles of ideal/real gas in the expansion from P_1, V_1, T_1 to V_2 . List-II includes change in various thermodynamic properties for the processes mentioned in List-I

	List-I		List-II
(I)	Isothermal reversible expansion of ideal gas	(P)	$\Delta S_{\text{sys}} = nR \ln \left(\frac{V_2}{V_1} \right)$
(II)	Isothermal irreversible expansion of ideal gas	(Q)	$\Delta S_{\text{total}} = 0$
(III)	Adiabatic free expansion of ideal gas	(R)	ΔS_{total} is +ve
(IV)	Adiabatic free expansion of real gas	(S)	$\Delta U = 0$
		(T)	$\Delta T \neq 0$
		(U)	$\Delta S_{\text{total}} = \Delta S_{\text{sys}}$

11. Among the following incorrect match is

- (A) II-P
- (B) I-Q
- (C) III-U
- (D) III-T

12. Among the following correct match is

- (A) IV-T
- (B) IV-P
- (C) III-Q
- (D) II-Q

SECTION-II : (Maximum Marks: 18)

- This section contains **SIX (06)** questions. The answer to each question is a **NUMERICAL VALUE**.
- For each question, enter the correct numerical value of the answer in the place designated to enter the answer. If the numerical value has more than two decimal places, **truncate/round-off** the value to **Two** decimal places; e.g. 6.25, 7.00, -0.33, -0.30, 30.27, -127.30, if answer is 11.36777..... then both 11.36 and 11.37 will be correct)
- Answer to each question will be evaluated according to the following marking scheme:

Full Marks : +3 If ONLY the correct numerical value is entered.

Zero Marks : 0 In all other cases.

1. 1.34g of organic compound with the molecular weight 134 gives 7.05g of AgI precipitate in the Ziesels method of estimation of methoxy groups. What is the weight percentage of methoxy groups in the organic compound?
(At wt of Ag is 108, I is 127)
2. Total number of stereo isomers possible for the octahedral complex $[M(AB)C_2de]$ are (Where : (AB) is unsymmetrical bi dentate ligand; c, d and e are mono dentate ligands)
3. In the reaction between permanganate and thiosulphate ions in faintly alkaline solutions, rate of disappearance of permanganate ion is how many times the rate of disappearance of thiosulphate ion?
4. When three moles of helium is mixed with one mole of neon at constant temperature and pressure (25°C , 1 atm), the entropy of mixing (in JK^{-1} , rounded off to two decimal place) is _____
(Given: $R = 8.314 \text{ JK}^{-1} \text{ mol}^{-1}$)
5. In the mixture $1.0 \times 10^{-3} \text{ M AgNO}_3 + 0.20 \text{ M NH}_3$, concentration of $[\text{Ag}(\text{NH}_3)]^+$ in correct scientific notation is $P \times 10^{-P} \text{ M}$. The value of 'P' is
Given : $[\text{Ag}(\text{NH}_3)_2]^+ \rightleftharpoons [\text{Ag}(\text{NH}_3)]^+ + \text{NH}_3; K_1 = 1.4 \times 10^{-6} \text{ M}$
 $[\text{Ag}(\text{NH}_3)]^+ \rightleftharpoons \text{Ag}^+ + \text{NH}_3; K_2 = 4.3 \times 10^{-4} \text{ M}$
6. Third virial coefficient for a real gas is $1.6 \times 10^{-3} \text{ dm}^6 / \text{mole}^2$ critical volume of that real gas (in $\text{cm}^3 / \text{mole}$) is

PART-3 : MATHEMATICS
SECTION-I (i) : (Maximum Marks: 32)

- This section contains **EIGHT (08)** questions.
- Each question has **FOUR** options. **ONE OR MORE THAN ONE** of these four option(s) is (are) correct answer(s).
- For each question, choose the option(s) corresponding to (all) the correct answer(s)
- Answer to each question will be evaluated according to the following marking scheme:

Full Marks : +4 If only (all) the correct option(s) is (are) chosen.

Partial Marks : +3 If all the four options are correct but **ONLY** three options are chosen.

Partial Marks : +2 If three or more options are correct but **ONLY** two options are chosen and both of which are correct.

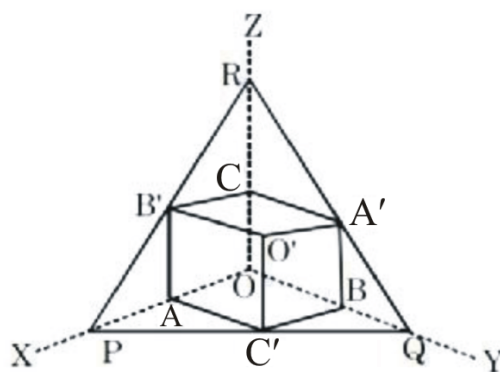
Partial Marks : +1 If two or more options are correct but **ONLY** one option is chosen and it is a correct option.

Zero Marks : 0 If none of the options is chosen (i.e. the question is unanswered).

Negative Marks : -1 In all other cases.

- **For Example :** If first, third and fourth are the **ONLY** three correct options for a question with second option being an incorrect option; selecting only all the three correct options will result in +4 marks. Selecting only two of the three correct options (e.g. the first and fourth options), without selecting any incorrect option (second option in this case), will result in +2 marks. Selecting only one of the three correct options (either first or third or fourth option), without selecting any incorrect option (second option in this case), will result in +1 marks. Selecting any incorrect option(s) (second option in this case), with or without selection of any correct option(s) will result in -1 marks.

1. Consider a parallelepiped $OAC'BA'CB'O'$ where $OA = 2$, $OB = 3$ and $OC = 4$, if areas of $\triangle OPQ$, $\triangle OQR$ and $\triangle OPR$ are least (as shown in figure), then



- (A) Volume of tetrahedron $OPQR$ is 24 cubic units
(B) Volume of tetrahedron $OPQR$ is 32 cubic units
(C) Shortest distance between BA' & PR is 3
(D) Shortest distance between BA' & PR is 6

2. If the sum to infinite terms of the series $\frac{1}{1.3.5.7.9} + \frac{1}{3.5.7.9.11} + \frac{1}{5.7.9.11.13} + \dots$ is 'k', then

(A) 'k' is rational number

(B) $k = \frac{1}{840}$

(C) $k = \frac{1}{842}$

(D) $\frac{1}{k}$ is a multiple of 2

3. The equation
$$\begin{vmatrix} (1+x)^2 & (1+x)^2 & -(2+x^3) \\ 2x+1 & 3x & 1-5x \\ x+1 & 2x & 2-3x \end{vmatrix} + \begin{vmatrix} (1+x)^2 & 2x+1 & x+1 \\ (1-x)^2 & 3x & 2x \\ 1-2x & 3x-2 & 2x-3 \end{vmatrix} = 0$$
, has

(A) No real solution

(B) Exactly 4 real solutions

(C) Exactly two real and two non-real solutions

(D) Infinite number of solutions, real or non-real

4. Let $X = \{1, 2, 3, 5, 6\}$ and a, c are natural numbers selected from set X with replacement.

Let $f(x) = \begin{cases} x^2 + x & x \geq 1 \\ x^2 - ax & x > 1 \end{cases}$, $g(x) = \begin{cases} x^2 - cx, & x \leq 2 \\ x^2 - 4x, & x > 2 \end{cases}$ and $h(x) = f(x).g(x)$, $x \in \mathbb{R}$

N_1 = Number of function h(x) such that f(x) and g(x) are both discontinuous

N_2 = Number of function h(x) such that h(x) is continuous at $x = 1$ and discontinuous at $x = 2$ but f(x) and g(x) are both discontinuous

N_3 = Number of function h(x) such that h(x) is discontinuous at $x = 1$ and continuous at $x = 2$, Then

(A) $N_1 = 30$

(B) $N_2 = 6$

(C) $N_3 = 10$

(D) $N_1 + N_2 + N_3 = 45$

5. Let $g(x) = \int_0^x (3t^2 + 2t + 9) dt$ and $f(x)$ be a decreasing function, $\forall x \geq 0$ such that $\overline{AB} = f(x)\hat{i} + g(x)\hat{j}$ and $\overline{AC} = g(x)\hat{i} + f(x)\hat{j}$ are the two smallest sides of $\triangle ABC$ whose circumcenter lies outside the triangle, $\forall x > 0$ then
- (A) $f(x) < 0$
- (B) $g(x) < 0$
- (C) $g(x) > 0$
- (D) $\lim_{t \rightarrow 0} \lim_{x \rightarrow \infty} \left(\cot\left(\frac{\pi}{4}(1-t^2)\right) \right)^{f(x)g(x)} = e$
6. A and B throw alternately throw a pair of dice. A wins if he throws a sum 6 before B throws 7 and B wins if he throws a 7 before A throws 6. If A starts the game, then
- (A) Probability that A wins is $\frac{31}{61}$
- (B) Probability that B wins is $\frac{31}{61}$
- (C) Probability that A wins is $\frac{30}{61}$
- (D) Probability that B wins is $\frac{30}{61}$
7. Let $f(x) = \cos^2 x \cdot e^{\tan x}$. $\forall x \in \left(-\frac{\pi}{2}, \frac{\pi}{2}\right)$ then
- (A) $f'(x)$ has a point of local minimum at $x = \frac{\pi}{4}$
- (B) $f'(x)$ has a point of local maximum in $\left(-\frac{\pi}{4}, 0\right)$
- (C) $f'(x)$ has exactly two points of local maximum/minimum in $\left(-\frac{\pi}{2}, \frac{\pi}{2}\right)$
- (D) $f''(x) = 0$ has no root in $\left(-\frac{\pi}{2}, \frac{\pi}{2}\right)$
8. The equation $10x^4 - 3x^2 - 1 = 0$ has
- (A) Atleast one root in $(-1, 0)$
- (B) Atleast one root in $(0, 1)$
- (C) Atleast two root in $(-1, 1)$
- (D) No root in $(-1, 1)$

SECTION-I (ii) : (Maximum Marks: 12)

- This section contains **TWO (02)** List-Match sets.
- Each List-Match set has **TWO** Multiple Choice Questions.
- Each List-Match set has two lists : **List-I** and **List-II**
- **List-I** has Four entries (I), (II), (III) and (IV) and **List-II** has **Six** entries (P), (Q), (R), (S), (T) and (U)
- **Four** options are given in each Multiple Choice Question based on List-I and List-II and **ONLY ONE** of these four options satisfies the condition asked in the Multiple Choice Question.
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Full Marks : +3 If ONLY the option corresponding to the correct combination is chosen.
 Zero Marks : 0 If none of the options is chosen (i.e., the question is unanswered)
 Negative Marks : -1 In all other cases

Paragraph for Questions 9 and 10

Answer Q.9 and Q.10 by appropriately matching the lists based on the information given in the paragraph.

	List-I		List-II
(I)	Let $f: \mathbb{R} \rightarrow \mathbb{R}$ and $f_n(x) = f(f_{n-1}(x)) \forall n \geq 2, n \in \mathbb{N}$, the roots of equation $f_3(x)f_2(x)f(x) - 25f_2(x)f(x) + 175f(x) = 375$ which also satisfy equation $f(x) = x$ will be	(P)	1
(II)	Let $f: [5, 10]$ onto $[4, 17]$, the integers in the range of $y = f(f(f(x)))$ is/are	(Q)	5
(III)	Let $f(x) = 8\cot^{-1}(\cot x) + 5\sin^{-1}(\sin x) + 4\tan^{-1}(\tan x) - \sin(\sin^{-1}x)$, then possible integral values which $f(x)$ can take	(R)	10
(IV)	Let 'a' denotes the roots of equation $\cos(\cos^{-1}x) + \sin^{-1} \sin\left(\frac{1+x^2}{2}\right) = 2\sec^{-1}(\sec x)$ then possible values of $[10a]$ where $[.]$ denotes the greatest integer function will be	(S)	15
		(T)	-3
		(U)	0

9. Which of following is correct combination?

- (A) (I) (P) (R) (B) (I) (Q) (R)
 (C) (II) (Q) (R) (S) (D) (II) (P) (S)

10. Which of following is correct combination?

- (A) (III) (P) (Q) (T) (U) (B) (III) (P) (Q) (R) (U)
 (C) (IV) (P) (R) (D) (IV) (Q) (S)

Paragraph for Questions 11 and 12

Answer Q.11 and Q.12 by appropriately matching the lists based on the information given in the paragraph

Let C be the set of curves having the property that the point of intersection of tangent with y -axis is equidistant the point of tangency and origin $(0, 0)$

	List-I		List-II
(I)	If $C_1, C_2 \in C$ C_1 : Curve is passing through $(1, 0)$ C_2 : Curve is passing through $(-1, 0)$ Then number of common tangents for C_1 and C_2	(P)	has a factor 2
(II)	If $C_3 \in C$, C_3 : is passing through $(2, 4)$. If $\frac{x}{a} + \frac{y}{b} = 1$ is tangent to C_3 , If $k_1a + k_2b^2 + k_3ab^2 = 0$, then $k_1 - k_2 + k_3$ (consider $k_1 - k_2 + k_3 > 0$)	(Q)	has a factor 3
(III)	If common tangents of C_1 and C_2 form an equilateral triangle, where $C_1, C_2 \in C$ and C_1 : curve passes through $(2, 0)$, if distance between centres of C_1 & C_2 is K then $3K$	(R)	less than 10
(IV)	If $C_1, C_2 \in C$ C_1 : Curve is passing through $(1, 0)$ C_2 : Curve is passing through $(-1, 0)$ Distance between the centres of the curves C_1 & C_2	(S)	greater than 5
		(T)	has factor 5
		(U)	has factor 6

11. Which of following is correct combination ?

- (A) (I) (Q) (S)
- (B) (I) (P) (R)
- (C) (II) (P) (S)
- (D) (II) (Q) (S)

12. Which of following is correct combination ?

- (A) (III) (P) (R)
- (B) (III) (P) (S)
- (C) (IV) (P) (S)
- (D) (IV) (Q) (S)

SECTION-II : (Maximum Marks: 18)

- This section contains **SIX (06)** questions. The answer to each question is a **NUMERICAL VALUE**.
- For each question, enter the correct numerical value of the answer in the place designated to enter the answer. If the numerical value has more than two decimal places, **truncate/round-off** the value to **Two** decimal places; e.g. 6.25, 7.00, -0.33, -.30, 30.27, -127.30, if answer is 11.36777..... then both 11.36 and 11.37 will be correct)
- Answer to each question will be evaluated according to the following marking scheme:

Full Marks : +3 If ONLY the correct numerical value is entered.

Zero Marks : 0 In all other cases.

- Let $f(x)$ be a differentiable function such that $f(f(x)) = x$, where $x \in (0, 1]$, also $f(0) = 1$, then if the value of $\int_0^1 (x - f(x))^{2020} dx$ is I then $\frac{1}{I}$ is
- $\lim_{n \rightarrow \infty} n^2 \int_0^{\frac{1}{n}} x^{2019x+1} dx$
- On circle $|z - 2i| = 2$; through origin 'O' a chord OP is drawn. On the tangent at O, a point T is taken so that $OT = OP$. If TP produced meet normal at O to the circle at Q, then limiting value of OQ as P moves towards origin is _____
(Note: P and T lie on the same side of imaginary axis)
- Let $f(x) = \cos^{-1} \left(\frac{2x}{1+x^2} \right)$, $g(x) = \cot^{-1} \left(\frac{2x}{x^2-1} \right)$ where $x \in (-1, 1)$. If area bounded by the curves $y = f(x) + g(x)$ and $y = \pi x^2$ is A then find the value of $[A]$
[Note: $[K]$ denotes greatest integer less than or equal to K.]
- If $\vec{a}, \vec{b}$ are vectors perpendicular to each other and $|\vec{a}| = 2, |\vec{b}| = 3, \vec{c} \times \vec{a} = \vec{b}$, then the least value of $2|\vec{c} - \vec{a}|$ is
- The shortest distance between origin and a point on the space curve $x = 2\sin t, y = 2\cos t, z = 3t$ is

Space for Rough Work

NAME OF THE CANDIDATE

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and shall abide by them.

Signature of the Candidate

I have verified the identity, name and Form
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